TinyTurla_Backdoor Write Up By Wh0isG0TH4M

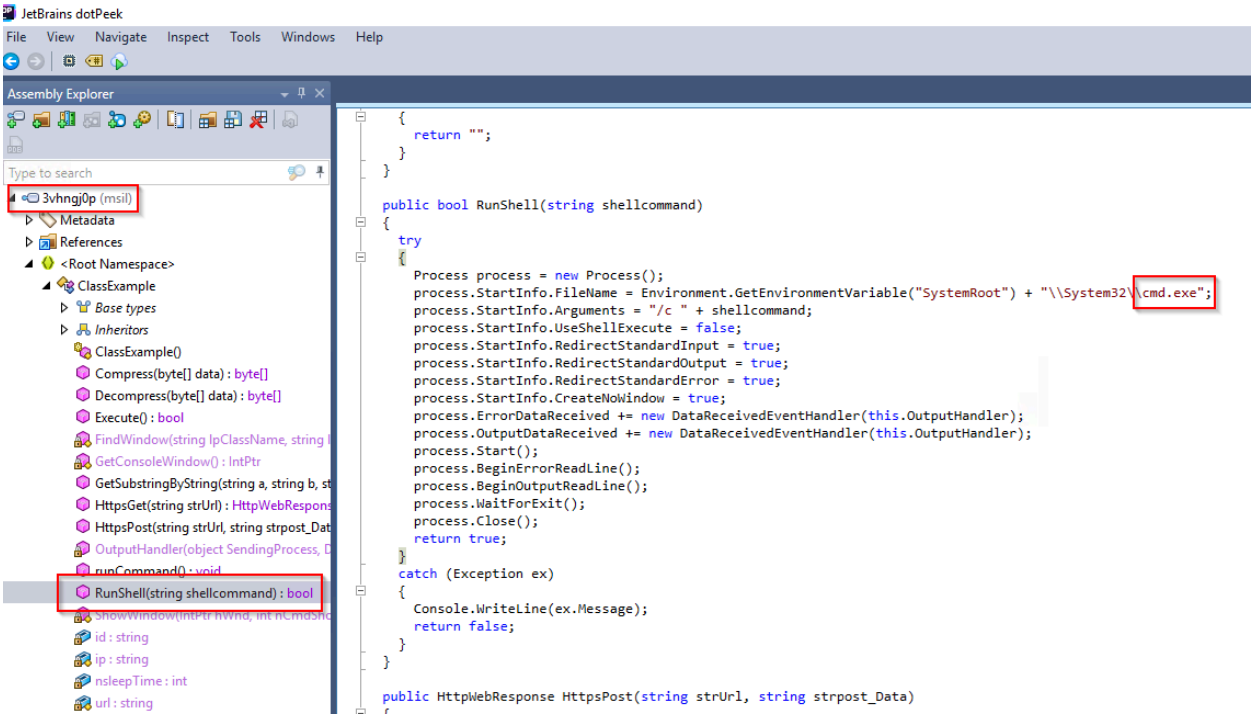
Sherlock Scenario

> You are a malware analyst assigned to investigate a suspected backdoor malware sample. The malware is designed to communicate with a remote server, execute various commands, and potentially exfiltrate data. Your task is to analyze the malware, understand its functionality, and determine its capabilities.

Difficulty: Medium

Task 0 : What is the name of the process used to run the shell command in the "RunShell" method?

Since DetectItEasy identified that the binary was .NET, I opened dotPeek to analyze the binary. dotPeek recognized the assembly name of the malicious binary as "**3vhngj0p**". Under the `RunShell` method, there was a process being spawned to execute the command.



Answer: cmd.exe

Task 1 : Which command in the "runCommand" method sets the sleep time for the program?

Inside `runCommand()`, the malware fetches, base64-decodes, and decompress incoming commands. If a command contains `[<sleep>]`, it updates its sleep duration variable (`this.nsleepTime`) using `int.Parse(subParm)` and sends back a base64-encoded confirmation: `"set sleep time ok."`



Answer: [<sleep>]

Task 2 : Which command in the "runCommand" method is used to execute a shell command?

The `[<shell>]` command executes because `runCommand()` iterates through the processed commands and checks if any line contains the `[<shell>]` tag using `c.Contains("[<shell>]")`. When a match is found, it spawns a new thread to run `this.RunShell(subParm)`



Answer: [<shell>]

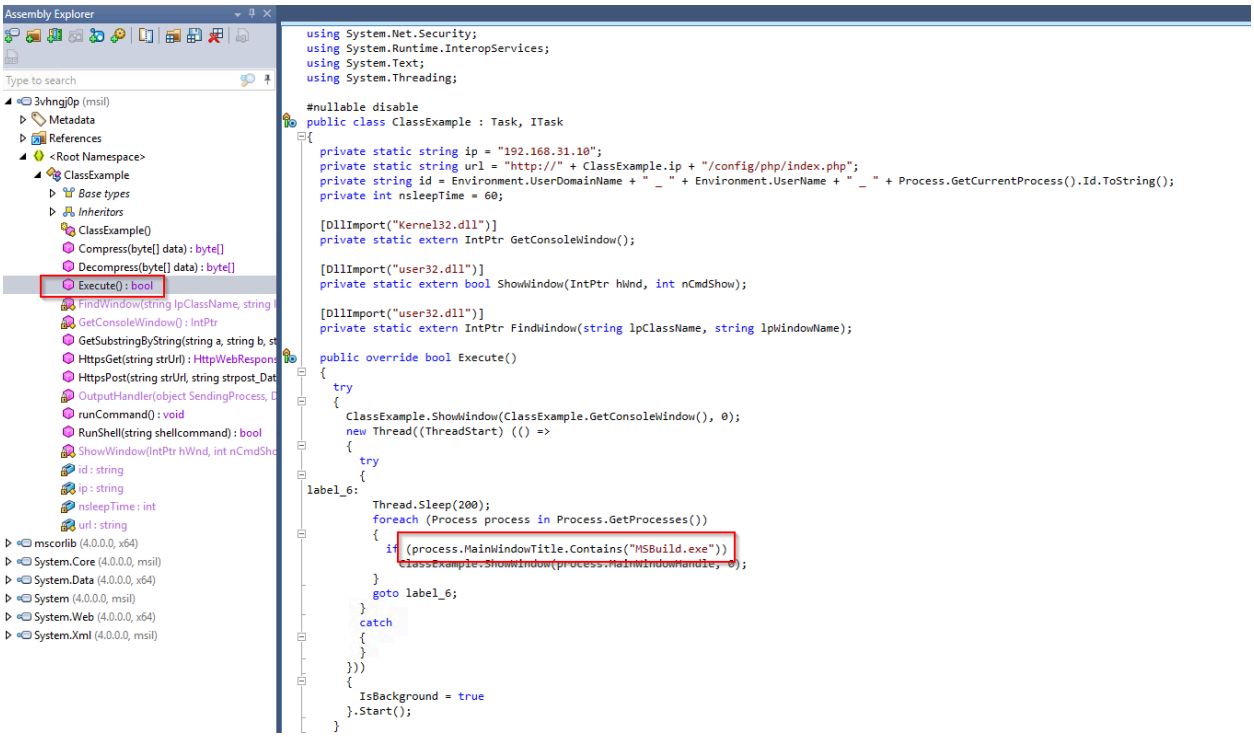
Task 3 : Which command in the "runCommand" method downloads a file to the server?

The `[<download>]` command executes because `runCommand()` checks if any line contains the `[<download>]` tag using `c.Contains("[<download>]")`. When matched, it spawns a new thread to read the file specified in `subParm`, compress and base64-encode its content, and send it out via an HTTP POST request.



Task 4 : Which process's main window title is specifically checked and hidden in the "Execute" method?

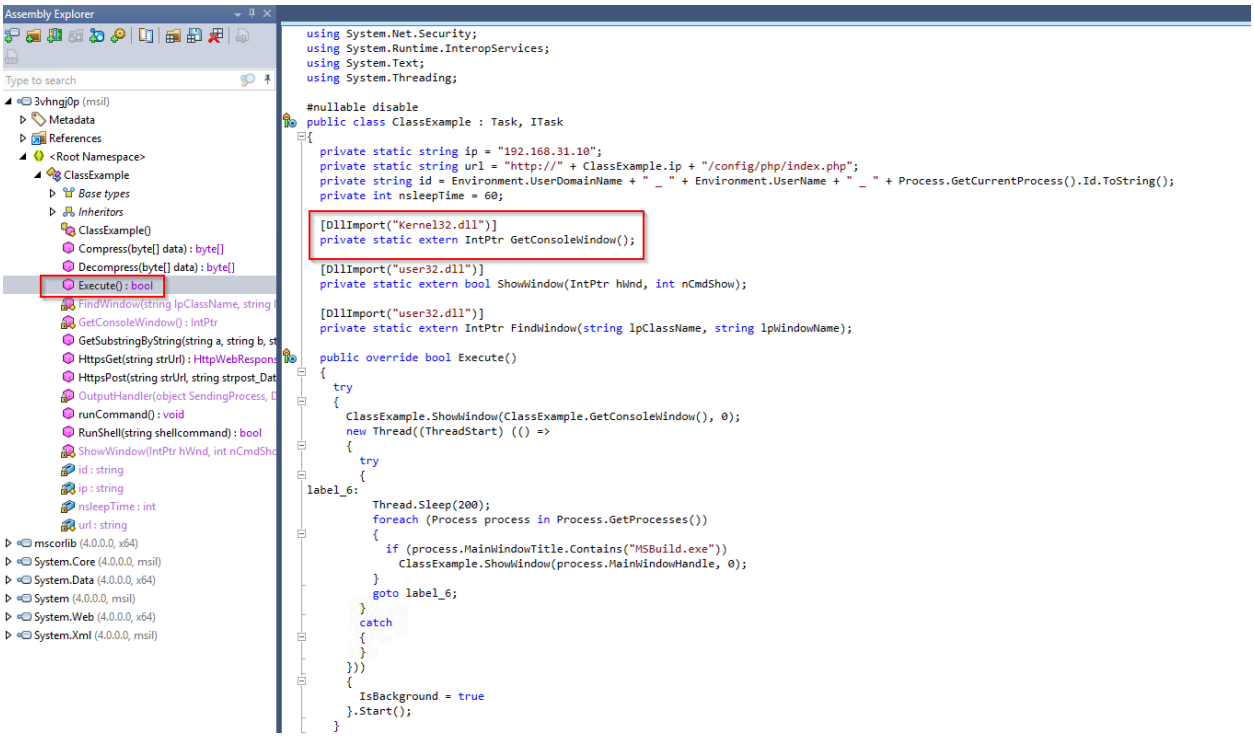
In the `Execute()` method of the `ClassExample` class, the malware iterates through active processes using `Process.GetProcesses()` and checks if `process.MainWindowTitle` contains `MSBuild.exe`. If found, it calls `ShowWindow(process.MainWindowHandle, 0)` to hide its window.



Answer: MSBuild.exe

Task 5 : Which DLL is imported to use the GetConsoleWindow function in the code?

In the `ClassExample` class, the malware uses `[DllImport("kernel32.dll")]` to import the native Windows API function `GetConsoleWindow()`, allowing it to retrieve the window handle of the current console process."



Answer: Kernel32.dll

Task 6 : What method is used to perform HTTP GET requests in the provided code?

HttpsGet method performs the GET request method



Answer: HttpsGet

Task 7 : Which IP address does the "HttpsPost" method use when making POST requests?

In the `ClassExample` class, the malware defines a hardcoded private static string variable `ip` initialized to `"192.168.31.10"`

```
#nullable disable
public class ClassExample : Task, ITask
{
  private static string ip = "192.168.31.10";
  private static string url = "http://" + ClassExample.ip + "/config/php/index.php";
  private string id = Environment.UserDomainName + " _ " + Environment.UserName + " _ " + P
  private int nsleepTime = 60;

  [DllImport("Kernel32.dll")]
  private static extern IntPtr GetConsoleWindow();

  [DllImport("user32.dll")]
  private static extern bool ShowWindow(IntPtr hWnd, int nCmdShow);

  [DllImport("user32.dll")]
  private static extern IntPtr FindWindow(string lpClassName, string lpWindowName);

  public override bool Execute()
```

Answer: 192.168.31.10

In conclusion, the TinyTurla Backdoor starts by executing on a Windows system. It checks running processes for a window title containing `MSBuild.exe` and hides the matching window. It then communicates with a remote server using HTTP GET

and POST requests to receive commands and send data. The malware decodes incoming Base64 data and decompresses the commands before processing them. The `[<sleep>]` command changes its sleep interval, while `[<shell>]` executes shell commands through `cmd.exe`. It can also use the `[<download>]` command to read a specified file, compress and Base64-encode its contents, and send the data to the remote server at `192.168.31.10`. This communication and command execution allow the malware to function as a backdoor, enabling remote control of the infected system and potentially allowing data exfiltration.